Inkloom AI

• ROUND 2: THE BUILD SUBMISSION

Intelligent Brand Governance Platform

A real-time, pre-publication review engine that validates draft copy against official brand voice guidelines, verified facts, and terminology policies before live publication.

TEAM DETAILS

SATYACODEZ

FutureStack Hackathon

TEAM LEADER

Satyam Kumar

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SOURCE CODE & BUILD

GitHub Verified

`satya-no17/futureStack`

Executive Summary: What Was Shipped in Round 2

From Architectural Blueprint to a Fully Functioning Multi-Tenant Product

01

4-Tier Hybrid Pipeline

Merged instant deterministic regex rules with semantic RAG retrieval and LLM claim verification. Checks words to avoid, brand voice tone, and technical claims.

Sub-50ms
DETERMINISTIC RULE SPEED

02

1-Click Intent Fixes

Generated in-place replacements that preserve the writer's original narrative and persuasive intent without generic destructive rewrites.

100%
OFFSET-GROUNDED FIXES

03

Human Review Loop

Complete editorial queue with visual character-level diff comparison, reviewer sign-offs, revision feedback notes, and publish gates.

4 Roles
RBAC PERMISSIONS MODEL

04

Multi-Tenant RAG

Strict tenant isolation powered by PostgreSQL pgvector. Pre-seeded with 3 enterprise, agency, and indie workspaces with distinct policies.

Zero Leak
CROSS-TENANT ISOLATION

Core AI Governance Engine: 4-Tier Pipeline

Multi-Dimensional Analysis Delivering Explainable Findings in Parallel

Tier 1 • Instant

Words to Avoid (Regex)

Scans for barred words, high-liability claims ("cheap", "guaranteed", "#1"), and outdated jargon using compiled word boundaries.

- Exact character start/end offsets
- Severity weights (High, Med, Low)
- Immediate preferred replacements

Tier 2 • Semantic RAG

Tone & Voice Alignment

Retrieves top-4 guideline vector chunks via cosine distance, querying Gemini to identify subtle tone deviations and hyperbole.

- 384-dimensional dense vectors
- Identifies sensationalism & slang
- Rule-cited explanations

Tier 3 • Fact Verification

Claim Grounding

Cross-examines claims against approved product technical specs to flag unsupported claims or contradicted statistics.

- Tri-state: Supported, Contradicted, Unverified
- Suppresses hallucinated claims
- Verifies SLAs & security certifications

Tier 4 • Readability

Accessibility & Scoring

Calculates Flesch-Kincaid grade level, sentence length distribution, and WCAG color contrast standards for distribution channels.

- Grade level target benchmarking
- Automated compliance score formula
- WCAG AA & AAA contrast ratios

INTERACTIVE UI

Draft Studio: Live Highlighter & 1-Click Fixes

Empowering Creators With Intent-Preserving Corrections, Not Destructive Rewrites

Interactive Visual Copy Highlighter

Draft text renders with non-overlapping visual highlight tokens corresponding to exact detected issues. Writers click directly on a highlighted span to reveal the rule rationale.

```
// Example live draft scan representation "Our crazy fast cloud is guaranteed to cut costs by 80% with instant setup." → Red Highlight: Banned Term ("crazy fast" → "sub-50ms median latency") → Yellow Highlight: Unverified Claim ("guaranteed" → "backed by 99.95% SLA") → Orange Highlight: Tone Violation ("instant" → "near real-time")
```

- Zero text jumping: Exact index offsets guarantee perfect caret and selection positioning.
- Direct edit mode: Creators can type modifications freely, then hit "Check My Words".

Intent-Preserving Correction Cards

Traditional AI proofreaders rewrite paragraphs destructively. Inkloom isolates the problematic phrase, identifies the exact brand rule broken, and delivers a 1-click replacement.

MUST FIX • BANNED TERM

Source: Brand Terminology Policy

Problem phrase: “cheap database hosting”

Diminishes brand perception of high-reliability infrastructure.

Fix: “cost-effective cloud storage” 1-Click Applied

- Instant score recalculation upon accepting suggestions.
- Optional skip with mandatory justification note for compliance tracking.

Collaborative Review Queue & Version Diff Viewer

Structured Human-in-the-Loop Governance for Distributed Marketing Teams

Dedicated Approval Queue

Submitted drafts automatically enter the Review Queue. Reviewers and Brand Leads can inspect the compliance score history, filter by channel, and take definitive action.

- **Approve:** Signs off on compliant drafts and moves status to "Approved".
- **Ask for Changes:** Sends constructive revision notes back to the author with highlighted problem sections.
- **Publish Gate:** Brand Leads can trigger final publication state once all checks pass.

100%
AUDIT LOGGED ACTIONS

0 Clicks
AUTOMATED HANDOFF

Side-by-Side Version Diff Viewer

Reviewers can immediately see exactly what words were added or removed between draft versions, eliminating manual proofreading time.

REMOVED FROM EARLIER DRAFT:

```
- "Our crazy fast platform is the cheapest on the market guaranteed."
```

ADDED IN REVISED DRAFT:

```
+ "Our platform delivers sub-50ms latency backed by our 99.95% uptime SLA."
```

Full version rollback and chronological thread commenting allow authors and approvers to align without leaving the platform.

Multi-Tenant Brand Rules & Fact Repositories

Hermetic Isolation Protecting Brand Equity Across Diverse Organizational Workspaces

Acme Enterprise Solutions

B2B Enterprise • Formal

Strict Fortune-500 compliance. Informal jargon, unverified latency claims, and unqualified guarantees are barred.

- 12 Banned Terms ("cheap", "hack", "crazy fast")
- 2 Verified Fact Sheets (SOC2 Type II, 99.95% SLA)
- Formal third-person voice enforcement

Zest Social App

Consumer Tech • Vibrant

Fast-paced Gen-Z social community. Dry corporate buzzwords are prohibited; energetic, clear messaging is enforced.

- 10 Avoided Terms ("synergy", "paradigm", "leverage")
- Verified Community Stats (500k active members)
- Active voice & casual, punchy headline rules

Solo Studio

Indie Creator • Technical

Autonomous developer workspace. Streamlined governance ensuring developer guides and technical newsletters are verified.

- Technical jargon checks ("magic", "no-brainer")
- Open source fact sheets & architecture docs
- Solo review & instant publish capabilities

Live Multi-Tenant Boundary Demonstration

The phrase “crazy fast hack” triggers 2 High-Severity violations in Acme, but passes compliant in Zest.

ISOLATED PGVECTOR PARTITIONS

Brand Health Analytics & Tamper-Evident Audit Trail

Transforming Editorial Corrections into Actionable Insights and Guideline Clarity

Organizational Score & Mistakes Breakdown

Aggregates scan performance across all authors, draft channels, and historical weeks to measure brand voice maturity.

91.5%
ORGANIZATION AVG SCORE

100%
RESOLVED CRITICAL ERRORS

Confusing Rules Helper
Identifies policies repeatedly violated by 3+ distinct team members. Highlights ambiguity so brand managers can rewrite confusing guidelines.

Tamper-Evident Chronological Audit Log

Every critical governance action is recorded in an immutable audit ledger containing actor ID, role, action type, entity ID, and metadata.

```
[02:11:36] Satyam Kumar scanned draft "Q4 Enterprise Platform"
[02:11:40] 1-Click fix applied on "crazy fast" → "sub-50ms"
[02:11:44] Sarah Approver approved draft v2 (Score: 100%)
[02:11:48] Victoria Brand authorized live publication
```

- Full accountability across distributed agencies and internal writers.
- Exportable compliance logs for corporate governance reviews.

Technology Stack & Implementation Deep Dive

High-Throughput, Serverless Cloud Database with Vector Grounding and Multi-Model AI

Frontend & Workflow

Next.js 16 + React 19 + Tailwind CSS

Built on Turbopack with custom Landing Page design system. Zero heavy UI dependencies, responsive glassmorphism navigation, and sub-100ms client state transitions.

```
- Next.js 16 App Router - Tailwind CSS v4 design tokens - Lucide React icon suite - Custom text offset highlighter
```

Backend API & Ingestion

Node.js + Express + TypeScript

Asynchronous pipeline executing deterministic rule checks and RAG retrieval concurrently. In-memory content hash caching prevents redundant LLM evaluations.

```
- Modular REST routing architecture - Zod typed schema validation - JWT & bcrypt RBAC middleware - SHA-256 payload memoization
```

Cloud DB & Vector Store

Neon Serverless PostgreSQL + pgvector

Serverless PostgreSQL cluster running pgvector. Performs sub-10ms cosine similarity searches across partitioned guideline and fact chunks.

```
- 384-dimensional dense vectors - HNSW indexed cosine distance - Automated schema migrations - Multi-tenant foreign keys
```

Google Gemini AI Integration Architecture

Uses `gemini-flash-lite-latest` with structured JSON outputs and automated fallback to `gemini-3.5-flash-lite`. Grounded by `gemini-embedding-001` vectors to eliminate hallucinations.

What Changed From Round 1 Proposal & Why

Strategic Engineering Decisions Made to Maximize Working Software Quality

1. Deep Text Governance over Heavy Vision Models

Architectural Focus

Original Plan: Equal distribution between computer vision image scanning and text governance.

What Changed: Market analysis showed 90% of compliance, legal, and tone risks originate in marketing copy, captions, and press claims. We prioritized sub-second text accuracy, 1-click fixes, and pgvector RAG, shipping an interactive WCAG contrast calculator for visual assets.

2. Transparent Error States vs Mock AI Fallback

Trust & Reliability

Original Plan: Built-in heuristic mock fallback if external LLMs experienced rate limits.

What Changed: In brand governance, fake AI responses destroy trust. We eliminated fallback simulations entirely, implementing multi-model failover and transparent UI banners: *"AI not responded: Rule checks only"* so compliance leads know exactly what was verified.

3. Multi-Tenant Side-by-Side Comparison Mode

Demo Innovation

Original Plan: Single workspace dashboard tested with sample text.

What Changed: To prove enterprise readiness, we pre-seeded and integrated 3 real-world workspaces (Enterprise, Social App, Indie Studio) with an interactive comparison screen proving strict tenant boundary enforcement.

Known Limitations & Commitments for Round 3

Three Substantial Product Milestones We Commit to Delivering for the Final Round

IMPROVEMENT 1

Multi-Modal Slide Deck Scanner

What it is: Native ingestion of multi-page PDF presentation decks and social image banners.

What it changes for the user: Marketing and sales teams can upload complete PowerPoint and Canva pitch decks to automatically verify logo safety margins, typography rules, and OCR-extracted factual claims in seconds.

IMPROVEMENT 2

1-Click CMS & Social Publishing

What it is: Live webhook and API distribution connectors for WordPress, Webflow, Ghost, LinkedIn, and Twitter/X.

What it changes for the user: Eliminates manual copy-pasting. Once a draft receives final reviewer approval in Inkloom, it automatically publishes or schedules directly into external production content channels.

IMPROVEMENT 3

Self-Healing Brand Rules Engine

What it is: Continuous AI clustering that monitors recurring reviewer overrides, author skips, and conflicting guidelines.

What it changes for the user: The system automatically suggests updated, clearer rule formulations to brand managers when multiple team members trip over ambiguous policies, keeping guidelines actively up to date.

Current Working Limitation: Slide deck OCR and external CMS publishing are currently simulated; scheduled for full Round 3 shipping.

TEAM SATYACODEZ COMMITTED